

3.8.2 Externally Marked Route X (PSV/EMPA)

The practical and investigative skills will be assessed through

- Practical Skills Verification (PSV) and
- Externally Marked Practical Assignment (EMPA).

The PSV requires teachers to verify their candidates' ability to demonstrate safe and skilful practical techniques and make valid and reliable observations

The EMPA has three stages where candidates

- undertake a themed task and collect data
- process the data
- complete a written EMPA test

Each stage must be carried out under controlled conditions but may be at a time convenient to the centre. The written test must be completed in a single uninterrupted session.

The EMPA is set and marked by AQA. Only one EMPA at each of AS and A2 will be provided in a given academic year. AQA will stipulate a period of time during which the EMPA (themed task and written test) must be completed.

Practical Skills Verification

Candidates following this route must undertake the practical activities outlined in sections 3.3 for AS and 3.6 for A2 in order to allow candidates suitable opportunities to demonstrate safe and skilful practical techniques and to make reliable and valid observations. The teacher will confirm, on the *Candidate Record Form*, for each candidate that this requirement has been met. Failure to complete the tick box will lead to a mark of zero being awarded to the candidate for the whole of this unit.

In the context of material specified in the relevant AS or A2 specification, candidates will be required to demonstrate the following skills

- following instructions
- selecting and using equipment
- organisation and safety.

Practical activities must provide candidates with opportunities to develop the knowledge and skills of *How Science Works* outlined in section 3.3 and 3.6. Teachers can provide the level of guidance that would normally be given during practical activities. However, in order to provide appropriate opportunities to demonstrate the necessary skills, instructions provided must not be too prescriptive but should allow candidates to make decisions for themselves, particularly concerning the organisation and conduct of practical work, and the manner in which equipment is used.

Further guidance for conducting practical activities for the PSV will be provided in the Teacher Resource Bank.

ICT

Candidates may use ICT where appropriate in the course of developing practical skills, for example in collecting and analysing data.

Externally Marked Practical Assignment (EMPA)

The Externally Marked Practical Assignment carries 50 marks and has two stages.

Stage 1: Themed task and collection of data

Candidates carry out practical work following AQA task sheets. The tasks may be conducted in a normal timetabled lesson and at a time convenient to the centre but must be under controlled conditions.

For AS, candidates collect raw data and represent it in a table of their own design on the *Candidate Result Sheet*. The candidates' work must be handed to the teacher at the end of each session.

For A2, candidates collect raw data on the *Candidate Result Sheet*. The candidates' work must be handed to the teacher at the end of each session.

Centres may use the task sheets, as described, or may make minor suitable modifications to materials or equipment following AQA guidelines. Any modifications made to the task sheet must be agreed with the Assessment Adviser and details must be provided to the AQA examiner. The task may be conducted in a normal timetabled session.

There is no specified time limit for this stage.

Stage 2: Processing of data

The teacher returns the candidates' data from Stage 1 (on the *Candidate Results Sheet*).

For AS, the teacher instructs the candidates to process the data (e.g. calculate means or rates of reaction) and plot an appropriate graph. The teacher must not instruct the candidates on the presentation of the data or on the choice of graph or chart.

For A2, the teacher instructs the candidates to process the data and carry out a suitable statistical test. The teacher must not instruct the candidates on the choice, implementation and interpretation of the statistical test.

For both AS and A2, stage 2 may be done in normal lesson time and must be done in a single session under controlled conditions. Both the raw and the processed data must be handed to the teacher at the end of the session.

Stage 1 and Stage 2 may be done in the same session. There is no specified time limit for Stage 2.

Stage 3: The EMPA written test

The EMPA written test should be taken as soon as convenient after completion of the Stage 2 and under controlled conditions. Each candidate is provided with an EMPA written test and the candidate's completed material from Stages 1 and 2.

The EMPA test is in two Sections.

Section A

This consists of a number of questions relating to the candidate's own data.

Section B

At the start of this section, candidates are supplied with additional data on a related topic. A number of questions relating to analysis and evaluation of the data then follow.

The number of marks allocated to each section may vary with each EMPA test.

Use of ICT during the EMPA

ICT may be used during the EMPA Stages 1 and 2 but teachers should note any restrictions in the Teachers' Notes. Use of the internet is not permitted.

Candidates absent for the practical work

A candidate absent for the practical work (Stage 1) should be given an opportunity to carry out the practical work before they sit the EMPA test. This may be with another group or at a different time. In exceptional cases, when this is not possible, the teacher can supply a candidate with class data. This must be noted on the CRF. In this case the candidate cannot be awarded marks for Stage 1, collection of data, but can still be awarded marks for Stage 2, processing of data.

Material from AQA

For each EMPA, AQA will provide:

- Teachers' Notes
- Task sheet
- EMPA test.

When received, this material must be kept under secure conditions. If it is to be used in more than one session, the centre must ensure security of material between sessions. Further details regarding this material will be provided.

Security of assignments

Completed EMPAs should be treated like examination papers and kept under secure conditions until sent to the AQA examiner.

General Information

Route X

Administration

Only one EMPA will be available in any year at AS and at A2. AQA will stipulate a period of time during which

the EMPA (themed task and written test) must be completed.

Candidates may make only one attempt at a particular EMPA and redrafting is **not** permitted at any stage during the EMPA.

Only Internally Assessed Units from the Biology specification can contribute towards an AS or A Level Award in Biology. **Human Biology Units cannot be used towards a Biology qualification.**

Work to be submitted

The material to be submitted to the AQA examiner for each candidate consists of

- the candidate's data from Stages 1 and 2 (on the *Candidate Result Sheet*)
- the EMPA written test and
- a *Candidate Record Form* including the PSV verification that safe and skilful practical techniques have been demonstrated and reliable and valid observations made.

In addition each centre must provide

- *Centre Declaration Sheet*
- Details of any agreed amendments to the task sheet must be notified to the AQA examiner with information supporting the changes from the Assessment Adviser.

Working in groups

For the PSV, candidates may work in groups provided that any skills being verified are the work of individual candidates. For the EMPA further guidance will be provided but the opportunity for group work will not be a common feature.

Other information

Section 6 of this specification outlines further guidance on the supervision and authentication of Internally assessed units.

Further support

AQA supports centres in a number of ways.

- A Teacher Resource Bank which includes further information and guidance
- Assessment Advisers appointed by AQA to provide advice on internally assessed units. Every centre is allocated an Assessment Adviser.

The Assessment Advisers can provide guidance on issues relating to the carrying out of tasks for assessment.

Any amendments to the EMPA task sheet must be discussed with the AQA Assessment Adviser and confirmation of the amendments must be submitted to the AQA examiner.